

CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 2: Polynomials

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the zeroes of $x^2 + 7x + 10$ and verify the relationship between the zeroes and the coefficients.
2. Find the zeroes of $4x^2 - 4x + 1$.
3. Find the zeroes of $5x^2 - 20$.
4. Find a quadratic polynomial whose zeroes have sum 4 and product -5 .
5. Find a quadratic polynomial whose zeroes are $\frac{2}{3}$ and $-\frac{1}{2}$.
6. If α and β are the zeroes of $x^2 - 6x + 8$, find $\alpha^2 + \beta^2$.
7. If α and β are the zeroes of $3x^2 + 5x - 2$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.
8. If one zero of $x^2 - 8x + k$ is 6, find k and the other zero.
9. If the sum of the zeroes of $kx^2 + 2x + 3k$ is equal to their product, find k .
10. The zeroes of $x^3 - 4x^2 + x + 6$ are $-1, 2$ and 3 . Verify the three relations between the zeroes and the coefficients.

Answer Key

1. -2 and -5 ; sum $= -7$, product $= 10$

2. $\frac{1}{2}$ and $\frac{1}{2}$

3. 2 and -2

4. $x^2 - 4x - 5$

5. $6x^2 - x - 2$

6. $36 - 16 = 20$

7. $\frac{-5/3}{-2/3} = \frac{5}{2}$

8. $k = 12$; the other zero is 2

9. $-\frac{2}{k} = 3$, so $k = -\frac{2}{3}$

10. Sum $= 4$; sum of products in pairs $= -2 + 6 - 3 = 1$; product $= -6$